

PROJECT REPORT ON

IoT-Based Smart Battery Health and End-of-Life Prediction System

Design, Simulation, Battery-State Estimation and Telemetry Analytics
using ESP32, INA219, DS18B20 and Wokwi

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August 2026

Educational / research prototype — simulation-based validation

CERTIFICATE

This is to certify that the project entitled “**IoT-Based Smart Battery Health and End-of-Life Prediction System**” is an original embedded-systems and battery-monitoring project carried out by **Shanmukh Patnala**. The work develops a simulated Battery Management System around an ESP32, integrating an INA219-compatible current/voltage monitor, DS18B20 temperature sensing, battery-state estimation, protection logic, telemetry generation and Python-based analytics.

The project demonstrates practical understanding of embedded sensing, I²C communication, battery-state estimation, coulomb counting, capacity degradation modeling, fault classification, data logging and remaining-useful-life estimation. Validation was performed using Wokwi simulation and recorded telemetry.

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DECLARATION

I hereby declare that the project entitled **“IoT-Based Smart Battery Health and End-of-Life Prediction System”** is an original work carried out as part of my self-learning and professional development in embedded systems, IoT, battery monitoring and data-driven engineering.

The implementation uses ESP32 firmware, a custom INA219-compatible Wokwi model, DS18B20 temperature sensing, deterministic SoC/SoH/RUL models and Python utilities for serial logging and analytics. The reported measurements and validation results are based on the supplied simulation evidence and demonstration dataset.

Place: Andhra Pradesh, India

Date: August 2026

Signature

SHANMUKH PATNALA

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported the development of this project. The work provided hands-on experience in embedded firmware, I²C sensor interfacing, battery monitoring, simulation, telemetry processing and predictive analytics.

I am thankful to the open-source hardware and software communities, Wokwi simulation resources, sensor documentation and programming libraries that made it possible to build and validate the prototype in a reproducible environment.

I also acknowledge the guidance of faculty members, mentors, peers and technical resources that contributed to the design, debugging, documentation and final presentation of the system.

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ABSTRACT

The IoT-Based Smart Battery Health and End-of-Life Prediction System is an ESP32-centered battery monitoring prototype designed to observe electrical and thermal operating conditions and convert them into useful battery-health indicators. An INA219-compatible I²C monitor provides bus voltage and load current, while a DS18B20 digital sensor measures temperature. The firmware calculates electrical power, State of Charge (SoC), State of Health (SoH), accumulated charge, equivalent full-cycle count, available capacity and Remaining Useful Life (RUL).

The embedded BMS uses a 2600 mAh nominal battery model and a 400-cycle nominal life model. SoC is estimated from the simulated cell voltage range of 3.00–4.20 V. SoH is modeled as capacity fade from 100% toward an 80% end-of-life threshold over 400 equivalent cycles. A protection layer classifies under-voltage, over-voltage, over-current and over-temperature conditions.

Telemetry is emitted in a CSV-friendly serial format for later storage and analysis. Python utilities are included for serial-to-CSV logging and degradation-based RUL estimation. The complete project is reproducible in Wokwi and includes the firmware, custom INA219 model, circuit definition, documentation, diagrams, sample data and validation screenshots.

Keywords: ESP32, Battery Management System, INA219, DS18B20, I²C, SoC, SoH, RUL, Coulomb Counting, Wokwi, IoT, Python, Battery Analytics.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

A Battery Management System is responsible for observing the electrical, thermal and health-related conditions of a battery and providing information that can be used for safe operation and maintenance. For IoT devices and embedded systems, a compact monitoring solution must acquire sensor data efficiently, derive meaningful battery indicators and expose the data in a form suitable for analysis.

This project develops a simulation-ready smart battery monitoring system around an ESP32. The design combines INA219 electrical measurement, DS18B20 temperature sensing, battery-state calculations, threshold-based safety classification and structured telemetry. The system is intentionally modular so that the simulation layer can later be replaced or extended with real battery hardware and a more sophisticated electrochemical or ML model.

1.2 Objectives

- Measure battery-side voltage, load current and temperature.
- Calculate electrical power from voltage and current.
- Estimate State of Charge using a voltage-based demonstration model.
- Estimate State of Health from a deterministic capacity-fade model.
- Integrate current over time to obtain accumulated charge and equivalent cycle count.
- Estimate remaining useful life in equivalent cycles.
- Detect under-voltage, over-voltage, over-current and over-temperature conditions.
- Generate machine-readable telemetry for CSV logging and Python analytics.
- Deliver a reproducible Wokwi project with source files, diagrams and validation evidence.

1.3 Problem Statement

Battery monitoring becomes difficult when raw electrical measurements must be converted into actionable health information. The design challenge is to combine sensing, state estimation, fault detection and data logging without making the embedded firmware unnecessarily complex. This project addresses that challenge with a compact ESP32-based architecture and deterministic models suitable for simulation and demonstration.

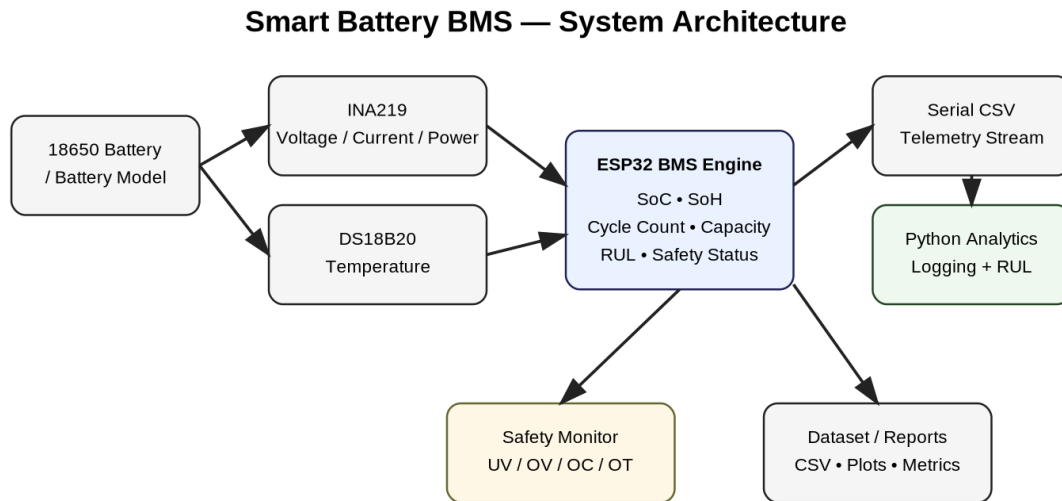
1.4 Scope

The scope covers embedded firmware, sensor interfacing, Wokwi simulation, battery-state estimation, protection logic, serial telemetry, sample-data analytics and documentation. The current implementation is a research/educational prototype. It does not claim production-grade battery protection, calibrated electrochemical SoC estimation or field-validated RUL prediction.

CHAPTER 2: PROPOSED SYSTEM / ARCHITECTURE

2.1 System Overview

The system is organized into five logical layers: the simulated battery and sensing layer, the ESP32 acquisition layer, the BMS estimation/protection layer, the telemetry layer and the Python analytics layer.



Educational simulation architecture — not a production battery protection system

Figure 2.1: Complete embedded-to-analytics system architecture.

2.2 Hardware and Simulation Subsystem

The simulated hardware uses an ESP32 DevKit-class controller, a custom INA219-compatible sensor model at I²C address 0x40 and a DS18B20 temperature sensor connected to GPIO4. A 4.7 kΩ pull-up is used on the DS18B20 data line. The Wokwi circuit provides a repeatable environment in which voltage, current and temperature conditions can be changed without requiring physical battery hardware.

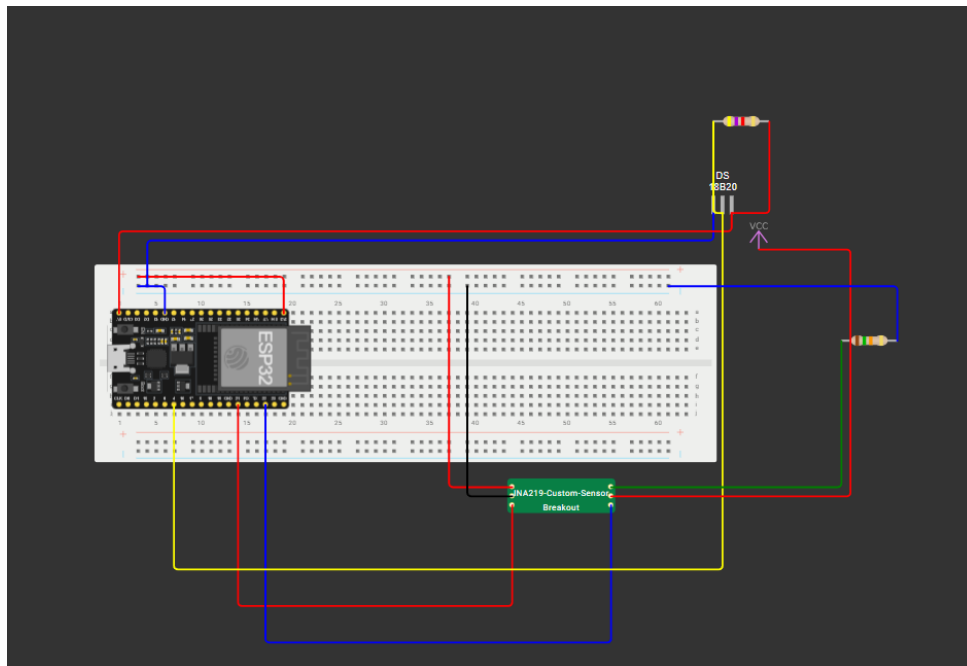


Figure 2.2: Wokwi hardware layout showing ESP32, custom INA219 sensor and DS18B20.

2.3 Sensor Interfaces

Element	Interface / Pin	Role
ESP32	Main controller	Acquisition, estimation, protection and telemetry
INA219	I ² C, address 0x40	Voltage, current and electrical power measurement
DS18B20	GPIO4 / 1-Wire	Temperature measurement
Serial output	115200 baud	Human-readable status and CSV-style telemetry

2.4 Data and Analytics Architecture

After every measurement cycle, the ESP32 updates battery state and periodically prints a detailed BMS status block followed by a machine-readable DATA record. The DATA record can be captured by the Python logger and stored as CSV. The RUL predictor then uses cycle count and SoH history to estimate the cycle at which the 80% SoH end-of-life threshold is reached.

CHAPTER 3: WORKING PRINCIPLE

3.1 Measurement Sequence

- Initialize I²C, INA219 and DS18B20 interfaces.
- Sample the INA219 bus voltage, shunt voltage and current.
- Add the shunt drop to obtain the simulated battery-side load voltage.
- Calculate power as voltage × current.
- Request and read the DS18B20 temperature.
- Integrate current over elapsed time for accumulated mAh.
- Update SoC, SoH, equivalent cycles, capacity and RUL.
- Evaluate safety thresholds and assign the current operating status.
- Print detailed status and CSV-friendly telemetry.

3.2 State of Charge Estimation

For the demonstration model, SoC is derived linearly from the simulated single-cell voltage range. With $V_{min} = 3.00\text{ V}$ and $V_{max} = 4.20\text{ V}$:

$$\text{SoC} = \text{clamp}(((V - 3.00) / 1.20) \times 100, 0, 100)$$

This is intentionally a simple educational model. A production BMS would normally use a calibrated OCV/SoC curve, current integration with correction, temperature compensation and a validated estimator.

3.3 Coulomb Counting and Equivalent Cycles

Assuming positive current represents battery discharge, accumulated charge is updated as:

$$\Delta Q \text{ (mAh)} = I \text{ (mA)} \times \Delta t \text{ (s)} / 3600$$

Equivalent full cycles are approximated by accumulated charge divided by the nominal capacity of 2600 mAh.

3.4 State of Health and Available Capacity

The simulation models capacity fade from 100% SoH at cycle 0 toward an 80% end-of-life threshold at 400 equivalent cycles. Available capacity is calculated as:

$$\text{Available Capacity} = 2600 \times \text{SoH} / 100 \text{ mAh}$$

3.5 Remaining Useful Life

$$\text{RUL} = \max(0, 400 - \text{equivalent cycle count})$$

CHAPTER 4: IMPLEMENTATION AND SPECIFICATIONS

4.1 Design Specifications

Parameter	Specification
Controller	ESP32 DevKit-class MCU
Electrical monitor	INA219-compatible I ² C model, address 0x40
Temperature sensor	DS18B20, GPIO4
Nominal battery capacity	2600 mAh
Nominal full voltage	4.20 V
Minimum voltage threshold	3.00 V
Maximum voltage threshold	4.20 V
Over-current threshold	1000 mA
Over-temperature threshold	45 °C
Nominal life model	400 equivalent full cycles
End-of-life SoH	80%
Sampling interval	500 ms
Detailed print interval	2 s
Serial baud rate	115200
Simulation environment	Wokwi

4.2 Embedded Software Organization

The firmware is organized around initialization, measurement, state update and reporting functions. Utility functions clamp calculated values, while dedicated functions calculate SoC, SoH, RUL and safety status. The BatteryStatus structure keeps all derived indicators together, which simplifies telemetry generation and future integration with data logging.

4.3 Custom INA219 Simulation Model

Because the project is intended to be reproducible in Wokwi, the repository contains a custom INA219-compatible chip model. The model exposes adjustable battery-voltage and load-current attributes and implements the key INA219 register reads used by the Adafruit library. This allows the firmware to exercise the I²C measurement path in simulation.

4.4 Python Analytics

The serial logger parses DATA records and appends timestamped rows to a CSV file. The RUL predictor reads cycle/SoH history, fits a linear degradation trend and estimates the cycle at which SoH reaches the 80% end-of-life threshold. The analytics implementation is deliberately lightweight and transparent so that the relationship between data and prediction can be inspected.

CHAPTER 5: RESULTS AND VERIFICATION

5.1 Verification Methodology

Validation was performed through Wokwi simulation and recorded serial telemetry. Tests cover normal operation and fault classification. The included evidence demonstrates that the system produces stable measurements, updates derived battery indicators and changes the status string when configured thresholds are exceeded.

5.2 Normal Operation

A representative normal-state observation is approximately 4.196 V, 150 mA, 629–630 mW and 22 °C. The corresponding calculated SoC is approximately 99.6%, SoH is 100%, available capacity is approximately 2600 mAh and the modeled remaining life is approximately 400 cycles.

```
=====
                BATTERY BMS STATUS
=====
Voltage       : 4.196 V
Current       : 150.00 mA
Power         : 630.00 mW
Temperature   : 22.00 C
SoC           : 99.60 %
SoH           : 100.00 %
Accumulated   : 0.6396 mAh
Cycle Count   : 0.000123
Available Capacity : 2600.00 mAh
Remaining Life : 400.00 cycles
STATUS        : NORMAL
=====
DATA,Voltage=4.196,Current=150.00,Power=630.00,Temperature=22.00,SoC=99.60,SoH=100.00,Cycles=0.000123,Capacity=2600.00,RUL=400.00,Status=NORMAL
```

Figure 5.1: Normal-operation BMS telemetry captured from the simulation.

5.3 Fault Detection – Over-Current

When the simulated load current is increased to approximately 1500 mA, the firmware detects that the 1000 mA threshold has been exceeded and reports **OVER_CURRENT**.

```
DATA,Voltage=4.196,Current=1500.00,Power=6294.00,Temperature=22.00,SoC=99.67,SoH=100.00,Cycles=0.00033,Capacity=2600.00,RUL=400.00,Status=OVER_CURRENT

----- BMS STATUS -----
Voltage       : 4.196 V
Current       : 1500.00 mA
Power         : 6294.00 mW
Temperature   : 22.00 C
SoC           : 99.67 %
SoH           : 100.00 %
Accumulated   : 1.7037 mAh
Cycle Count   : 0.00066
Available Cap. : 2600.00 mAh
Remaining Life : 400.00 cycles
STATUS        : OVER_CURRENT
-----

DATA,Voltage=4.196,Current=1500.00,Power=6294.00,Temperature=22.00,SoC=99.67,SoH=100.00,Cycles=0.00066,Capacity=2600.00,RUL=400.00,Status=OVER_CURRENT
```

Figure 5.2: Over-current validation at approximately 1500 mA.

5.4 Fault Detection – Over-Voltage

A simulated voltage of approximately 4.203 V is above the configured 4.20 V threshold. The BMS therefore reports **OVERVOLTAGE**, demonstrating threshold-based protection classification.

SERIAL MONITOR	CHIPS CONSOLE
----- BMS STATUS -----	
Voltage	: 4.203 V
Current	: 150.00 mA
Power	: 628.00 mW
Temperature	: 22.00 C
SoC	: 100.00 %
SoH	: 100.00 %
Accumulated	: 0.3538 mAh
Cycle Count	: 0.00014
Available Cap.	: 2600.00 mAh
Remaining Life	: 400.00 cycles
STATUS	: OVERVOLTAGE

DATA,Voltage=4.203,Current=150.00,Power=628.00,Temperature=22.00,SoC=100.00,Cycles=0.00014,Capacity=2600.00,RUL=400.00,Status=OVERVOLTAGE	

Figure 5.3: Over-voltage validation at approximately 4.203 V.

5.5 Validation Summary

Test / Condition	Observed Evidence	Result
Normal operation	4.196 V, 150 mA, 22 °C, STATUS=NORMAL	PASS
Over-current	1500 mA, STATUS=OVER_CURRENT	PASS
Over-voltage	4.203 V, STATUS=OVERVOLTAGE	PASS
CSV telemetry	DATA,... record generated	PASS
Cycle accumulation	Accumulated mAh and cycle count increase with time	PASS
SoH / capacity model	SoH and available capacity follow degradation model	PASS
RUL estimation	RUL reported in cycles	PASS

5.6 Telemetry Visualization

The supplied Wokwi plot demonstrates that the monitored signal changes over simulation time. The original plot is useful as a direct simulation artifact, while the additional dataset plots provide a clearer engineering view of degradation behavior.

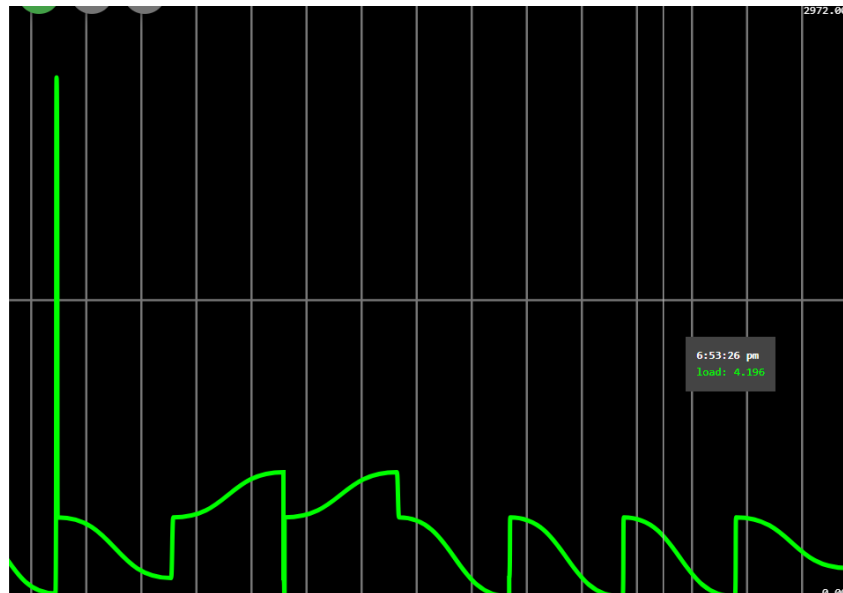


Figure 5.4: Supplied Wokwi telemetry plot.

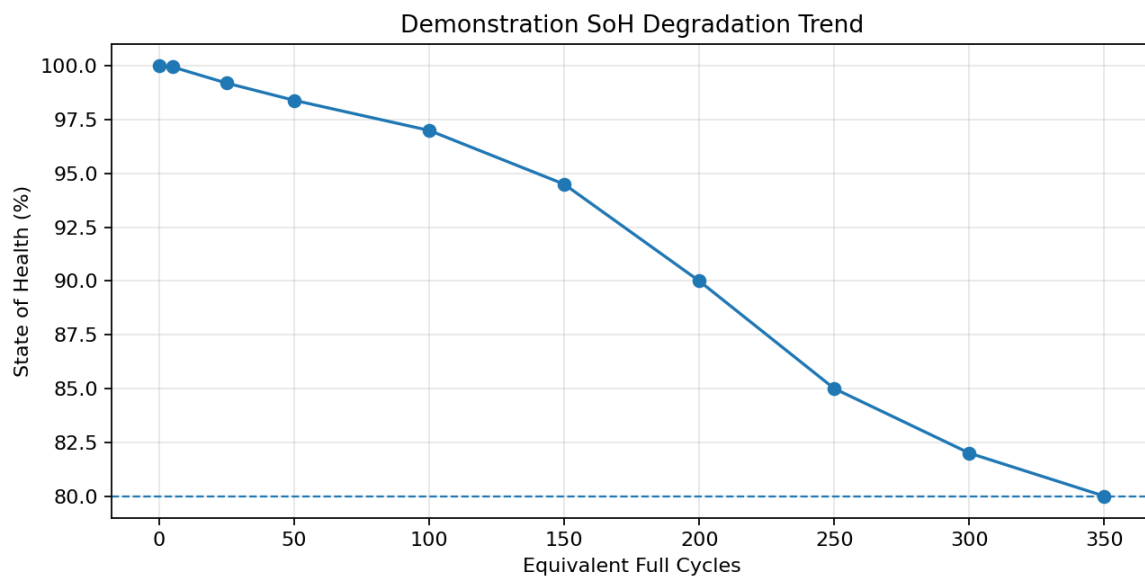


Figure 5.5: Clear SoH degradation trend generated from the included demonstration dataset.

5.7 Capacity Trend

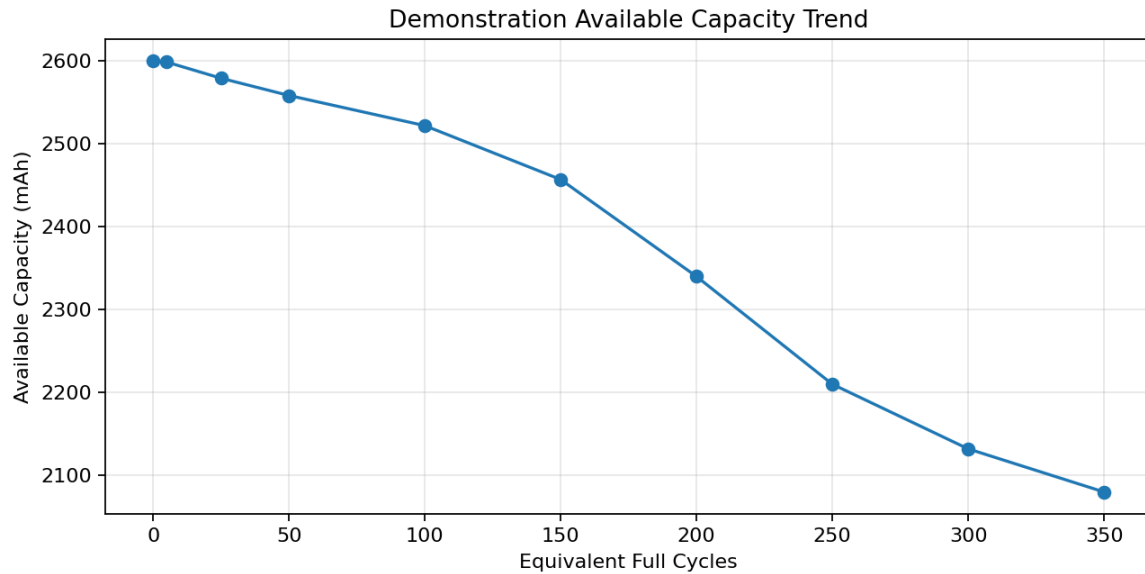


Figure 5.6: Demonstration available-capacity trend.

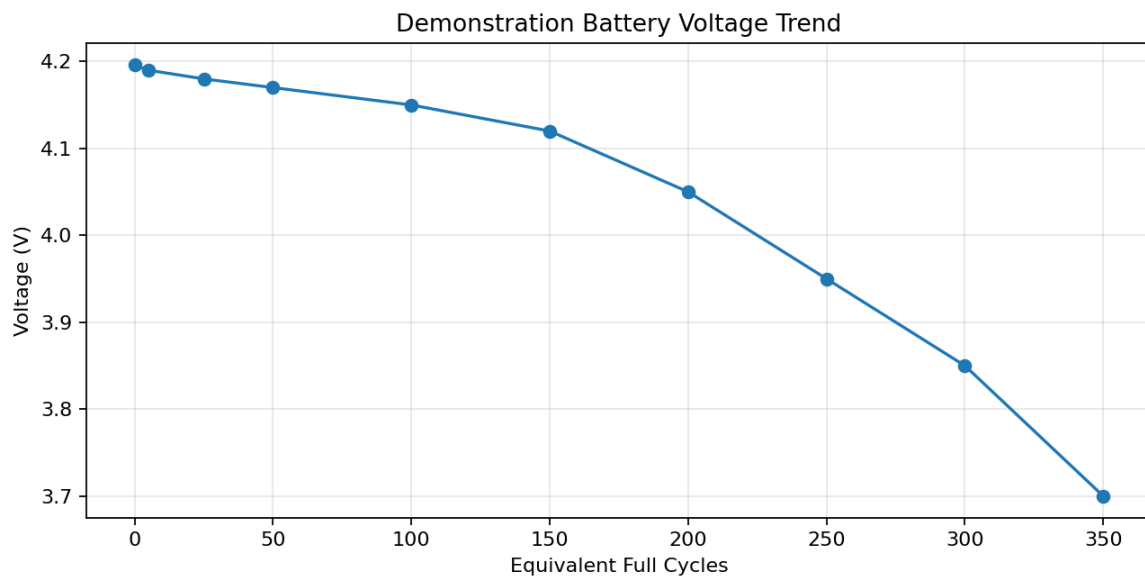


Figure 5.7: Demonstration voltage trend over equivalent cycles.

5.8 Demonstration Dataset

The included dataset contains ten representative points ranging from near-new operation to a degraded 80% SoH endpoint. At 350 equivalent cycles, the example data reports approximately 80% SoH, 2080 mAh available capacity and 50 cycles of modeled remaining life. These values are demonstration data and should not be interpreted as experimentally measured aging characteristics of a specific commercial 18650 cell.

CHAPTER 6: ADVANTAGES, LIMITATIONS AND APPLICATIONS

6.1 Advantages

- Combines electrical, thermal and battery-health indicators in one compact firmware flow.
- Uses standard I²C sensing concepts and a modular ESP32 architecture.
- Provides both human-readable diagnostics and machine-readable telemetry.
- Supports repeatable simulation without requiring physical battery hardware.
- Includes explicit safety-status classification for common threshold violations.
- Provides a clear path from embedded sensing to Python analytics.
- Repository structure is suitable for Git-based version control and future extension.

6.2 Limitations

- The present implementation is simulation-based; no physical battery validation is claimed.
- The SoC model is a simple linear voltage mapping and is not a calibrated electrochemical estimator.
- The SoH model is deterministic and assumes a simplified linear capacity-fade relationship.
- The RUL estimate is a model-based engineering estimate, not a field-validated prediction.
- The custom INA219 model reproduces the subset of registers required by the firmware rather than every production-device behavior.
- Protection logic reports status but does not itself disconnect a physical battery through dedicated safety hardware.

6.3 Applications

- IoT battery monitoring prototypes.
- Embedded-system and BMS education.
- Battery telemetry collection and dataset generation.
- Battery degradation and RUL algorithm experimentation.
- FPGA/MCU-based monitoring system demonstrations.
- Early-stage research before physical battery characterization.

CHAPTER 7: FUTURE SCOPE

- Connect the firmware to a real 18650 cell through appropriate protection and power-stage hardware.
- Add a calibrated OCV/SoC lookup table or a model-based estimator such as an equivalent-circuit observer.
- Replace the deterministic SoH model with experimentally measured aging data.
- Expand RUL prediction using validated regression, ensemble learning or sequence models.
- Add persistent storage and timestamp synchronization for long-duration field datasets.
- Add Wi-Fi/Bluetooth telemetry for remote dashboards and alerts.
- Introduce a web or mobile dashboard for live voltage, current, temperature, SoC and SoH.
- Add physical protection circuitry for over-current, over-voltage, under-voltage and thermal faults.
- Perform calibration, thermal validation, cycle testing and uncertainty analysis.
- Evaluate resource usage, latency and energy consumption on the ESP32.

7.1 Suggested Next Development Stage

The most valuable next stage is to collect a controlled real-battery dataset. The same CSV telemetry format can be retained, allowing the embedded system and analytics pipeline to evolve without changing the overall software architecture. A measured dataset would provide the basis for validating SoC, SoH and RUL models.

CHAPTER 8: CONCLUSION

The IoT-Based Smart Battery Health and End-of-Life Prediction System demonstrates a complete simulation-oriented battery monitoring workflow using an ESP32, INA219-compatible electrical sensing and DS18B20 temperature measurement. The firmware converts raw measurements into power, SoC, SoH, accumulated charge, equivalent cycles, available capacity and RUL indicators.

The project also demonstrates threshold-based protection classification and structured telemetry suitable for downstream data processing. Normal-operation, over-current and over-voltage simulations provide direct evidence that the safety-status logic responds to configured operating limits.

The accompanying Python logger and RUL estimator extend the system from embedded monitoring into an analytics workflow. Although the current models are intentionally simplified, the architecture is suitable for later replacement with calibrated battery models and experimentally validated machine-learning methods.

Overall, the project provides a reproducible foundation for learning and further development in embedded BMS design, IoT telemetry, battery analytics and predictive maintenance.

CHAPTER 9: REFERENCES

1. Texas Instruments, *INA219 26-V, 12-bit, I²C output current/voltage/power monitor*, device documentation.
2. Analog Devices / Maxim Integrated, *DS18B20 Programmable Resolution 1-Wire Digital Thermometer*, device documentation.
3. Espressif Systems, *ESP32 Series Documentation*, microcontroller and peripheral reference material.
4. Adafruit Industries, *Adafruit INA219 Arduino Library*, Arduino sensor interface library.
5. Wokwi, *Online Electronics Simulator*, used for reproducible virtual-hardware simulation.
6. Python Software Foundation, *Python Documentation*, programming language reference.
7. NumPy / pandas / Matplotlib documentation, used for dataset processing and visualization.
8. Project source files and validation artifacts included with this deliverable: ESP32 firmware, custom INA219 model, Wokwi diagram, sample dataset and Python analytics.

Engineering Note

All numerical battery-health and RUL values in this report are simulation or demonstration-model results. They must not be used as a substitute for a certified battery protection system, cell characterization, or a production BMS.

APPENDIX A: PROJECT FILE STRUCTURE

```
smart-battery-bms/  
■■■ sketch.ino  
■■■ diagram.json  
■■■ ina219.chip.c  
■■■ ina219.chip.json  
■■■ libraries.txt  
■■■ data/sample_battery_data.csv  
■■■ python/bms_logger.py  
■■■ python/rul_predictor.py  
■■■ docs/architecture.md  
■■■ docs/diagrams/*.svg  
■■■ docs/screenshots/*.png  
■■■ README.md  
■■■ LICENSE
```

APPENDIX B: REPRESENTATIVE SERIAL TELEMETRY

DATA, Voltage=4.196, Current=150.00, Power=629.40, Temperature=22.00, SoC=99.60, SoH=100.00, Cycles=0.000123, Capacity=2600.00, RUL=400.00, Status=NORMAL

APPENDIX C: SIMULATION THRESHOLDS

Parameter	Configured value	Fault / role
Minimum voltage	3.00 V	UNDER_VOLTAGE
Maximum voltage	4.20 V	OVER_VOLTAGE
Maximum current	1000 mA	OVER_CURRENT
Maximum temperature	45 °C	OVER_TEMPERATURE
Nominal capacity	2600 mAh	Model parameter
Nominal life	400 cycles	Model parameter
End-of-life SoH	80%	Model threshold